WARNING: Tube/Valve amplifiers use potentially LETHAL HIGH VOLTAGES.
Building, troubleshooting and testing of these amplifiers should only be
performed by someone who is thoroughly familiar with
the safety precautions around high voltages.

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Universal tube preamp PCB, ECC99/6N6P/ECC88/ECC85/6922/6DJ8 etc. MU follower or SRPP

Koifarm · 2022-07-15 10:31 am

Koifarm
Member

2022-07-15 10:31 am #1

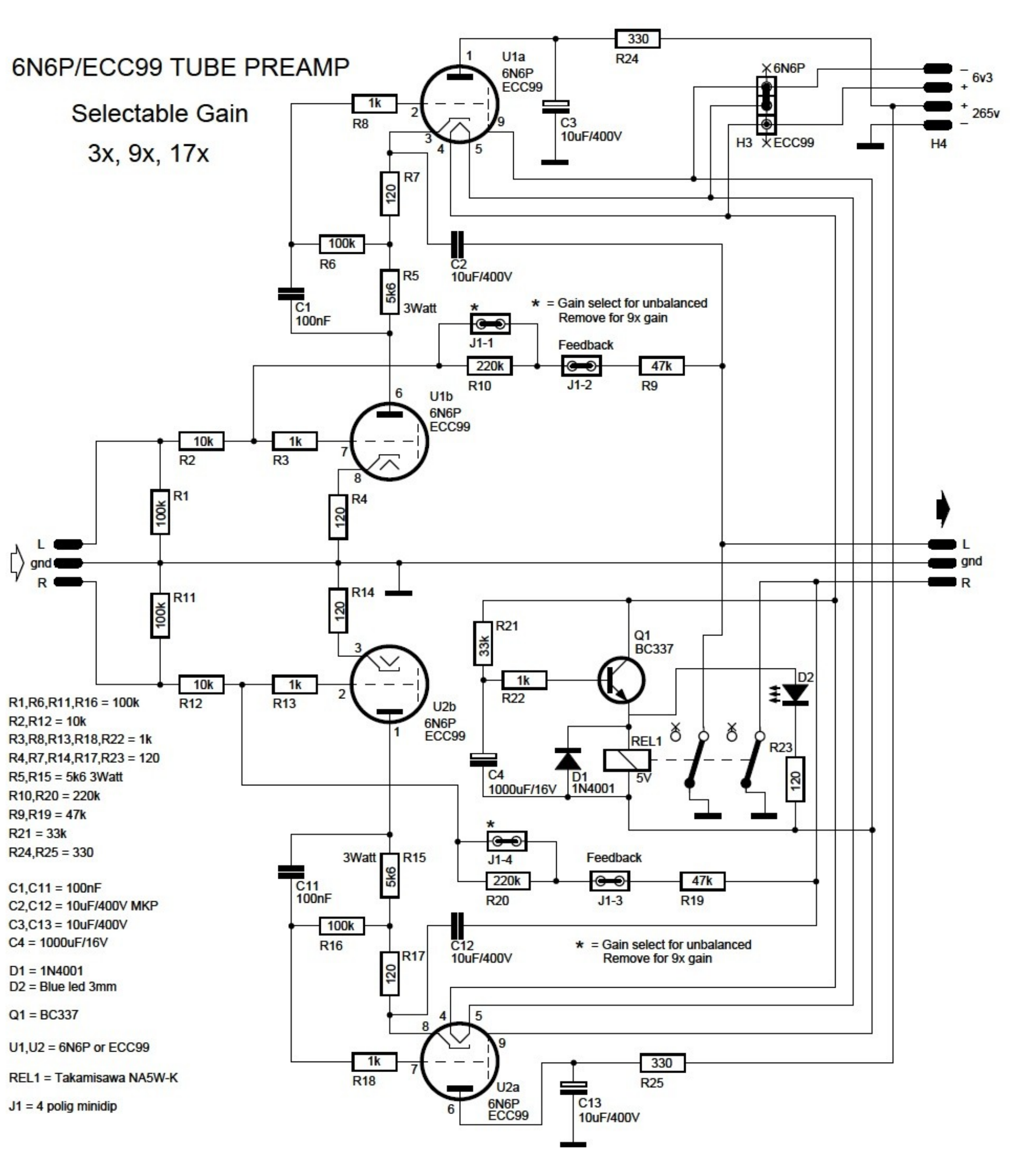
Hello DIYaudio community,

as spin-off of this thread : https://www.diyaudio.com/community/threads/universal-pcb-for-purifi-1et400a-amps.379211/ i have made a universal tube preamp PCB i like to share with you all.
This PCB is 100 x 100mm. And productioncost on JLCPCB are very low.

Features:

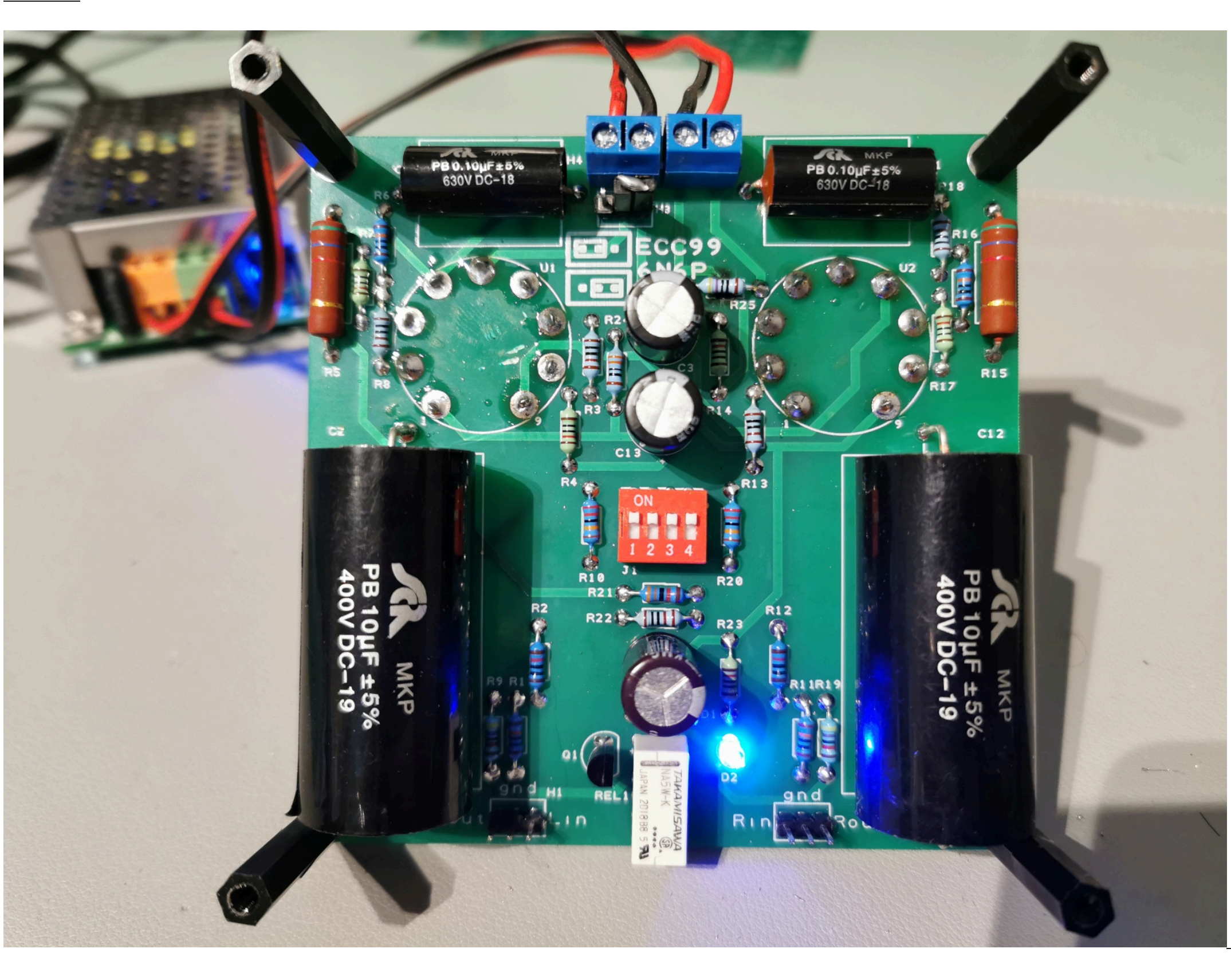
- 2 Channels
- Selectable gain = 3x, 9x, 17x
- Startup mute delay to prevent pop noise
- Selectable heater supply for ECC99 or 6N6P pinout
- Suitable output impedance 2k or higher.
- Low distortion, no humm, no hiss.
- Mu-Follower or SRPP topology.
- Also for ECC88, ECC85, 6922, 6DJ8 etc. tubes.
- SMPS Supply 265v and 6.3v.
- Balanced(XLR) possible with two PCB's.

Schematic

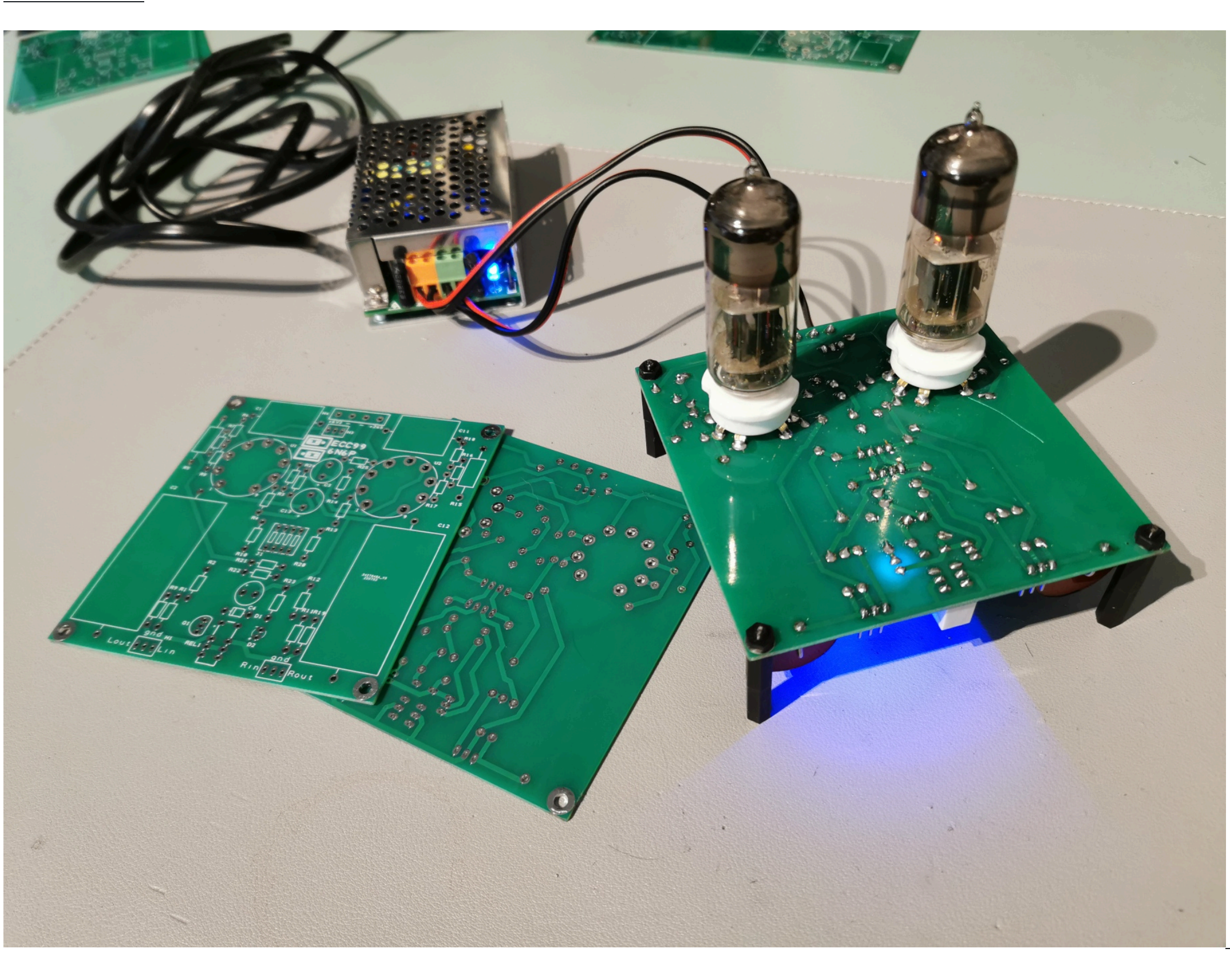


This is the MU-Follower tube preamp. Circuit around Q1(BC337) is the start-up delay who mutes the output for 20 seconds.
With minidip switch J1-1(J1-4 other channel) you can select gain 3x resp. 9x when J1-2(J1-3) is on(feedback).
When J1-2(J1-3) is off then there is no feedback and gain is 17x.
With jumper H3 you must select the right heater supply. For tubes with 6.3v paralell heaters to pin 9 select ECC99,
And for tubes with heatersupply 6.3v only on pin 4 and 5 select 6N6P.

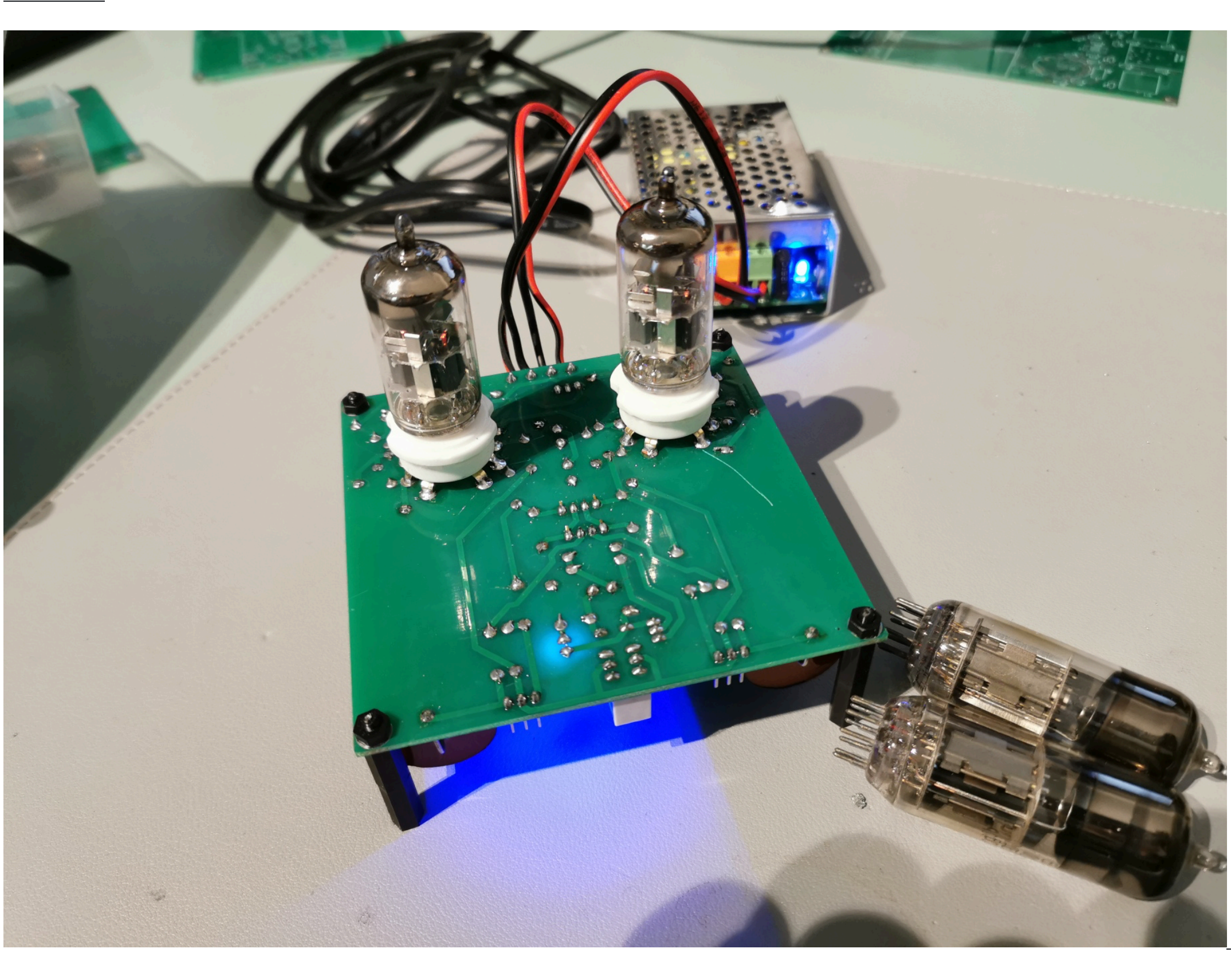
The build



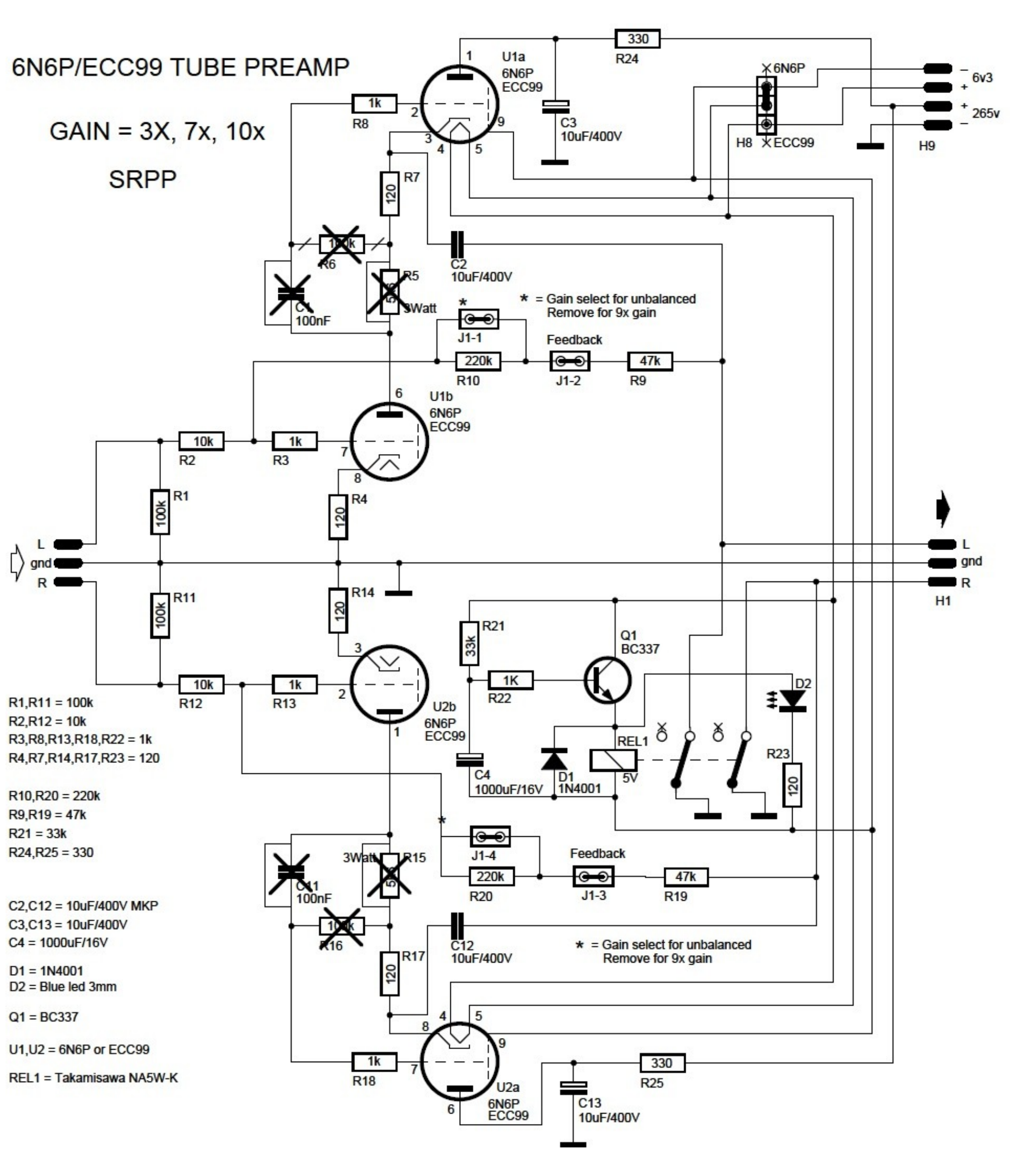
With 6N6P tubes



With ECC88



SRPP mod



To use the SRPP topology you have to remove R6 and R16, and bypass C1, C11, R5 and R15.
Gain = 3x, 7x or 10x(no feedback).

Powersupply

This is a SMPS from Aliexpress with 265V and 6.3V output.

PCB

Below are the gerberfiles to order your own PCB

Attachments

Gerber universal tube P...

Last edited: 2022-07-17 7:13 pm

weissi, Bas Horneman, Boydk and 4 others

Report Reply

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alain98
Member

2022-07-15 4:50 pm #2

Hy Koifarm,
Do you have a PDF file of your pcb to make my own pcb?
Thanks in advance
Alain

Report Reply

---

ruffrecords
Member

2022-07-15 4:50 pm #3

I think you should include a bypass capacitor for the lower triode.

Cheers

ian

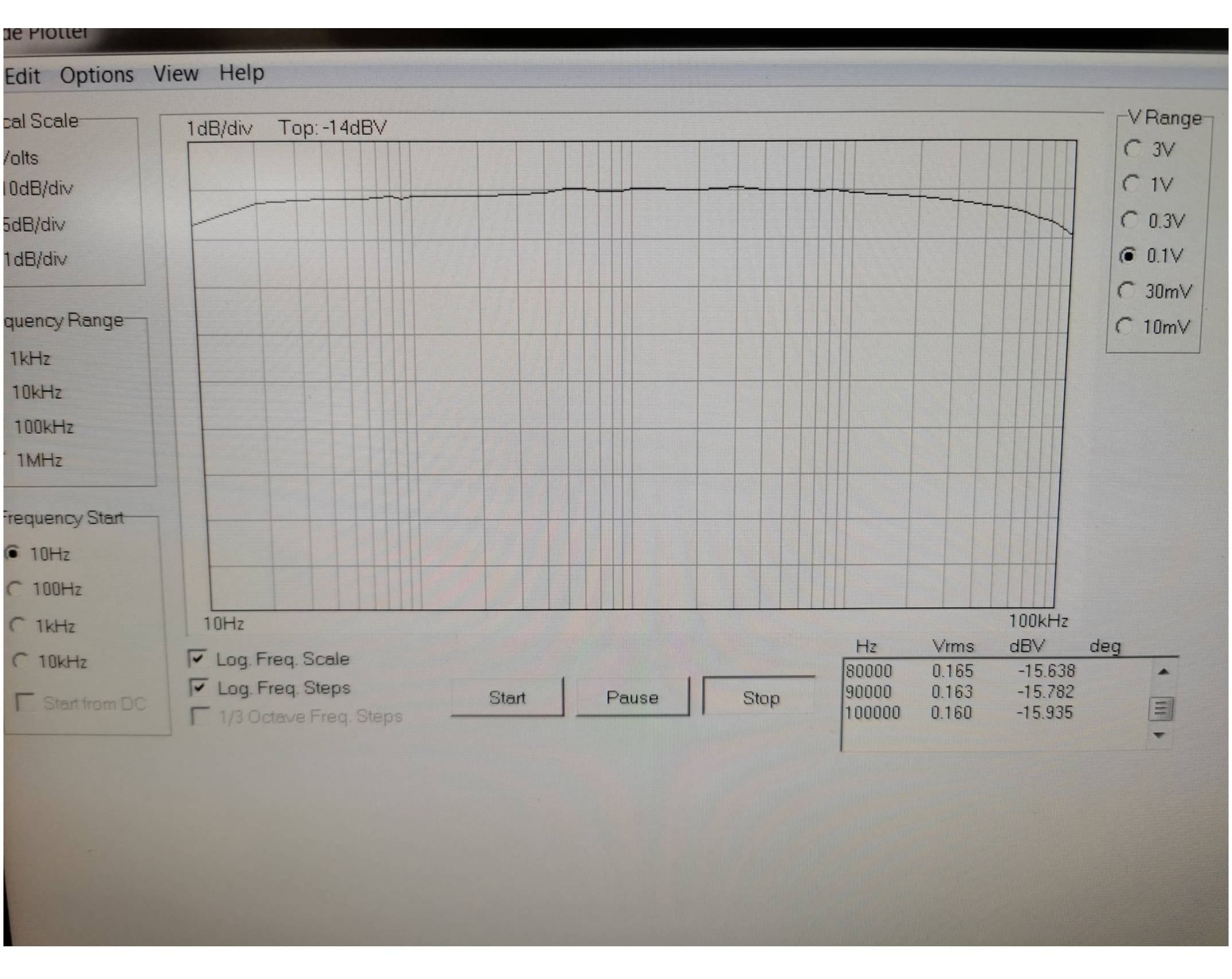
Report Reply

Koifarm
Member

2022-07-15 5:14 pm #4

Alain, i can generate the pdf files after my holiday.
There are gerber to pdf converters online.

Ian, frequency respons is from 10hz to 100khz within +/-0.5db and gain is enough.
But thanks for the reply.



Hz	Vrms	dBV	deg
80000	0.165	-15.638	
90000	0.163	-15.782	
100000	0.160	-15.935	

Last edited: 2022-07-15 5:22 pm

Report Reply

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rongon
Member

2022-07-15 5:42 pm #5

Local shunt NFB! Nice. That's a cool little circuit.
Do you have a link to the power supply you mentioned?

Report Reply

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rongon
Member

2022-07-15 5:48 pm #6

> ruffrecords said:
> I think you should include a bypass capacitor for the lower triode.
>
> Cheers
>
> ian

Perhaps sub an LED for R4 and R14?

Report Reply

Koifarm
Member

2022-07-15 5:48 pm #7

Link for smps:

https://nl.aliexpress.com/item/1005...216579079850164762ec166!12000029255708991!sea

https://www.aliexpress.com/item/1005001407107303.html?spm=a2g0o.detail.1000014.10.525f7cceWG2j7G&gps-id=pcDetailBottomMoreOtherSeller&scm=1007.40050.281175.0&scm_id=1007.40050.281175.0&scm-url=1007.40050.281175.0&pvid=7b953e52-5229-49f4-9564-a3facfdebdaa&_t=gps-id😛cDetailBottomMoreOtherSeller,scm-url:1007.40050.281175.0,pvid:7b953e52-5229-49f4-9564-a3facfdebdaa,tpp_buckets:668#2846#8115#2000&pdp_ext_f={"sku_id":"12000015963231639","sceneId":"30050"}&pdp_npi=2@dis!EUR!!45.69!!!!!@211b5db316551372258983283ee191!12000015963231639!rec

https://www.aliexpress.com/item/32997695785.html?spm=a2g0o.productlist.0.0.1cad7ea2fTpgxl&algo_pvid=4a3e6604-8832-4a24-a7f6-8790a3a5400e&algo_exp_id=4a3e6604-8832-4a24-a7f6-8790a3a5400e-32&pdp_ext_f={"sku_id":"66980528735"}&pdp_npi=2@dis!EUR!!29.79!!!!!@0b0a01f816551358878617330eb7a3!66980528735!sea

https://www.aliexpress.com/item/100...b16551378287617573ee8f6!12000018192145879!sea

Last edited: 2022-07-15 6:01 pm

Report Reply

Koifarm
Member

2022-07-15 5:50 pm #8

> rongon said:
> Perhaps sub an LED for R4 and R14?

As you see freq.respons is very linear and distorsion is 0.0043%
Noise floor is also -118db.

Report Reply

---

ruffrecords
Member

2022-07-15 7:24 pm #9

> Koifarm said:
> Ian, frequency respons is from 10hz to 100khz within +/-0.5db and gain is enough.

But open loop gain will be improved with bypassed cathode so distortion will be lower but a LED will probably do the trick since the PCB already exists.

Cheers

Ian

Report Reply

---

ruffrecords
Member

2022-07-15 7:27 pm #10

> Koifarm said:
> Noise floor is also -118db.

This is an amazingly good noise floor. . Can you show the actual measurement and the conditions under which they were made please?

Cheers

ian

Report Reply

Koifarm
Member

2022-07-15 7:37 pm #11

Ian, when i am back from holiday i shall make some measurements with different output impedance and topology. I use Arta and Rew software with Behringer UMC202.

With SMPS there is no 50/100hz interference, this helps.

Report Reply

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ruffrecords
Member

2022-07-15 9:41 pm #12

> Koifarm said:
> Ian, when i am back from holiday i shall make some measurements with different output impedance and topology. I use Arta and Rew software with Behringer UMC202.

One thing to be aware of with software like ARTA and REW is that their FFT displays usually show the noise per root Hertz. To get the rms noise (which is what you hear) you have to integrate over the 20KHz bandwidth. Assuming the noise is pretty flat then this just means adding about 43dB (20 log (root(20K))to the per root Hz noise floor. Edit: REW will show you the A or C weighted rms noise.

> Koifarm said:
> With SMPS there is no 50/100hz interference, this helps.

I agree, I use them all the time now for heaters. I recommend MeanWell,

Cheers

ian

rongon

Report Reply

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ruffrecords
Member

2022-07-15 10:12 pm #13

I wish someone would come up with an HT only SMPSU. The aliexpress ones all have 6.3V outputs and you **have** to use them because that is the one that is regulated and the HT is just rectified and smoothed.

Cheers

Ian

Report Reply

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alain98
Member

2022-07-16 12:27 pm #14

Koifarm, I used the software Gerbview to convert gerbers to pdf with success.
Thanks.

Koifarm

Report Reply

Koifarm
Member

2022-07-16 3:09 pm #15

Making your own PCB is nice. But for €7.50 you get 5 PCB incl. tax and shippingcost.

EL Reinoo

Report Reply

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dBel84
Member

2022-07-17 4:14 am #16

Hi. thank you for sharing your work. I have always liked the sound of an SRPP topology and this seems like a very simple project to try out. I used your gerbers to place an order for PCBs . I have never seen these compact SMPS units before either, I have used 6.3V smps for heater supplies but never for the B+.

I will let you know how I get along..dB

Koifarm

Report Reply

Koifarm
Member

2022-07-17 9:12 am #17

I can recomend these smps. No problems with EMI or hiss.

Report Reply

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jean-paul
diyAudio Moderator Emeritus

2022-07-17 9:19 am #18

Measured with RF capable equipment? All too often I read this but then my stuff does register other reality.

Report Reply

Koifarm
Member

2022-07-17 10:10 am #19

No measurement, just my experience with the phonostage who is nearby in wooden housing.

Report Reply

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jean-paul
diyAudio Moderator Emeritus

2022-07-17 10:12 am #20

As expected, sorry. It is often assumed one can not hear RF so it is not an issue. EMI and RF are nasty in audio though and in semiconductor stuff sometimes the cause of "listening fatigue" and this intermittent or depending on the time of the day/temp etc. Even when devices look nice and have blue LEDs 🙂 Not measuring = not knowing.

BTW I do like your design so no negative criticism except a few quibbles like lacking volume control and PSU. Nice design.

Last edited: 2022-07-17 10:22 am

Report Reply

Write your reply...

Attach files Post reply

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